

Lecture 2

Optimization and Linear Classifiers

Machine Learning I

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Outline

Gradient Descent: Convergence

The Perceptron

Support Vector Machines

Logistic Regression

The Loss Functions, Side by Side

Where We Are

Lecture 1 established the recipe — hypothesis class, loss, optimizer — and worked it out for squared loss, where the optimizer is a closed form.

Today the loss changes and the closed form disappears. Three questions:

1. When does gradient descent actually converge, and how fast? (§1)
2. What is the simplest learning algorithm for a linear classifier, and what can be proved about it? (§2)
3. Which convex losses stand in for the 0/1 loss, and what do their minimizers look like? (§3–4)

Throughout, labels are $y \in \{-1, +1\}$ for the perceptron and SVM, and $y \in \{0, 1\}$ for logistic regression. The two conventions are traditional; a slide near the end reconciles them.

Gradient Descent: Convergence

Smoothness

Gradient descent needs the gradient not to change too fast. The standard assumption:

L -smooth

$f: \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth if its gradient is L -Lipschitz:

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y}.$$

For twice-differentiable f this is equivalent to $\lambda_{\max}(\nabla^2 f(\mathbf{x})) \leq L$ for all $\mathbf{x}$.

- Least squares: $\nabla^2 \text{RSS} = 2\mathbf{X}^\top \mathbf{X}$, so $L = 2\lambda_{\max}(\mathbf{X}^\top \mathbf{X})$.
- Logistic loss (later today): $L \leq \frac{1}{4}\lambda_{\max}(\mathbf{X}^\top \mathbf{X})$.
- The 0/1 loss is not smooth — not even continuous. Nothing here applies to it.

The Descent Lemma

If f is L -smooth, then for all $\mathbf{x}, \mathbf{y}$: $f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2$

Proof: integrate the gradient along the segment

Let $g(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ for $t \in [0, 1]$, so $g'(t) = \nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))^\top (\mathbf{y} - \mathbf{x})$. Then

$$\begin{aligned} f(\mathbf{y}) - f(\mathbf{x}) &= g(1) - g(0) = \int_0^1 g'(t) dt = \int_0^1 \nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))^\top (\mathbf{y} - \mathbf{x}) dt \\ &= \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \int_0^1 [\nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - \nabla f(\mathbf{x})]^\top (\mathbf{y} - \mathbf{x}) dt \\ &\leq \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \int_0^1 \|\nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - \nabla f(\mathbf{x})\| \|\mathbf{y} - \mathbf{x}\| dt \\ &\leq \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \int_0^1 L t \|\mathbf{y} - \mathbf{x}\|^2 dt = \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2. \quad \square \end{aligned}$$

Third line: Cauchy–Schwarz. Fourth: L -smoothness. A smooth function sits below a quadratic that touches it at $\mathbf{x}$; gradient descent minimizes that quadratic.

One Step Decreases the Loss

Apply the lemma with $\mathbf{y} = \mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)$, writing $\mathbf{g}_t = \nabla f(\mathbf{x}_t)$:

Sufficient decrease

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) + \mathbf{g}_t^\top (-\eta \mathbf{g}_t) + \frac{L}{2} \|\eta \mathbf{g}_t\|^2 = f(\mathbf{x}_t) - \eta \left(1 - \frac{L\eta}{2}\right) \|\mathbf{g}_t\|^2.$$

With $\eta \leq 1/L$ the bracket is at least $\frac{1}{2}$, so

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \frac{\eta}{2} \|\nabla f(\mathbf{x}_t)\|^2$$

- The loss never increases, and it drops by an amount proportional to the squared gradient.
- Summing over t : $\frac{\eta}{2} \sum_t \|\mathbf{g}_t\|^2 \leq f(\mathbf{x}_0) - f^* < \infty$, so $\|\mathbf{g}_t\| \rightarrow 0$. Gradient descent finds a stationary point — for *any* smooth f , convex or not.
- The step $\eta = 1/L$ is the largest one this argument certifies. Lecture 1's $\eta < 2/a$ for the quadratic is the same fact with the exact constant.

Convexity

Convex

f is convex if for all $\mathbf{x}, \mathbf{y}$ and $\lambda \in [0, 1]$: $f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y})$.

For differentiable f , equivalently, f lies *above* every tangent:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) \quad \forall \mathbf{x}, \mathbf{y}.$$

For twice-differentiable f , equivalently, $\nabla^2 f(\mathbf{x}) \succeq 0$ everywhere.

Smooth: f is below a quadratic through the tangent. Convex: f is above the tangent.
Squeezed between the two, gradient descent cannot get lost.

Consequence of the tangent inequality at a minimizer $\mathbf{x}^*$: any stationary point is a global minimizer, because $\nabla f(\mathbf{x}) = \mathbf{0} \Rightarrow f(\mathbf{y}) \geq f(\mathbf{x}) \quad \forall \mathbf{y}$.

Convergence Rate for Convex Smooth Functions

Theorem

If f is convex and L -smooth, gradient descent with $\eta = 1/L$ satisfies, for every $t \geq 1$,

$$f(\mathbf{x}_t) - f^* \leq \frac{L \|\mathbf{x}_0 - \mathbf{x}^*\|^2}{2t}.$$

Proof, part 1: distance to $\mathbf{x}^*$ shrinks

By convexity at $\mathbf{x}_t$: $f^* = f(\mathbf{x}^*) \geq f(\mathbf{x}_t) + \mathbf{g}_t^\top (\mathbf{x}^* - \mathbf{x}_t)$, i.e. $f(\mathbf{x}_t) - f^* \leq \mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*)$. Combine with sufficient decrease ($\eta = 1/L$):

$$\begin{aligned} f(\mathbf{x}_{t+1}) - f^* &\leq f(\mathbf{x}_t) - f^* - \frac{1}{2L} \|\mathbf{g}_t\|^2 \leq \mathbf{g}_t^\top (\mathbf{x}_t - \mathbf{x}^*) - \frac{1}{2L} \|\mathbf{g}_t\|^2 \\ &= \frac{L}{2} \left(\|\mathbf{x}_t - \mathbf{x}^*\|^2 - \left\| \mathbf{x}_t - \mathbf{x}^* - \frac{1}{L} \mathbf{g}_t \right\|^2 \right) = \frac{L}{2} \left(\|\mathbf{x}_t - \mathbf{x}^*\|^2 - \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \right), \end{aligned}$$

where the middle equality is the identity $\|\mathbf{a}\|^2 - \|\mathbf{a} - \mathbf{b}\|^2 = 2\mathbf{a}^\top \mathbf{b} - \|\mathbf{b}\|^2$ with $\mathbf{a} = \mathbf{x}_t - \mathbf{x}^*$, $\mathbf{b} = \frac{1}{L} \mathbf{g}_t$.

Convergence Rate: Finishing the Proof

Proof, part 2: telescope

Sum the last inequality over $s = 0, \dots, t-1$. The right side telescopes:

$$\sum_{s=0}^{t-1} (f(\mathbf{x}_{s+1}) - f^*) \leq \frac{L}{2} (\|\mathbf{x}_0 - \mathbf{x}^*\|^2 - \|\mathbf{x}_t - \mathbf{x}^*\|^2) \leq \frac{L}{2} \|\mathbf{x}_0 - \mathbf{x}^*\|^2.$$

Since $f(\mathbf{x}_s)$ is non-increasing, each of the t terms on the left is at least the last one:

$$t(f(\mathbf{x}_t) - f^*) \leq \sum_{s=0}^{t-1} (f(\mathbf{x}_{s+1}) - f^*) \leq \frac{L}{2} \|\mathbf{x}_0 - \mathbf{x}^*\|^2. \quad \square$$

- Rate $O(1/t)$: to get within ϵ of the optimum takes $O(1/\epsilon)$ iterations.
- If in addition f is μ -strongly convex ($\nabla^2 f \succeq \mu \mathbf{I}$), the rate improves to $\|\mathbf{x}_t - \mathbf{x}^*\|^2 \leq (1 - \mu/L)^t \|\mathbf{x}_0 - \mathbf{x}^*\|^2$: linear convergence, governed by the condition number L/μ . (Proof in the homework.)

In Practice

Step size

- L is rarely known. Try $\eta \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$; the loss curve tells you.
- Rising loss $\Rightarrow \eta$ too large. Flat loss $\Rightarrow \eta$ too small.
- Standardize features first: it lowers L/μ and makes one η work for all coordinates.

Stopping

- $\|\nabla f(\mathbf{x}_t)\| < \epsilon$, or
- relative change $|f(\mathbf{x}_t) - f(\mathbf{x}_{t-1})| / |f(\mathbf{x}_t)| < \epsilon$, or
- validation error stops improving (early stopping — a regularizer, Lecture 3).

For SGD the sufficient-decrease bound fails step by step, but holds *in expectation*; the same $O(1/t)$ analysis goes through with a decaying $\eta_t \propto 1/\sqrt{t}$.

The Perceptron

Linear Classifiers

For $y \in \{-1, +1\}$, a linear classifier predicts

$$f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x} + b).$$

- The decision boundary $\{\mathbf{x} : \mathbf{w}^\top \mathbf{x} + b = 0\}$ is a hyperplane with normal vector $\mathbf{w}$.
- An example is classified correctly iff $y(\mathbf{w}^\top \mathbf{x} + b) > 0$.
- Absorb b as before: $\mathbf{x} \leftarrow (\mathbf{x}, 1)$, $\mathbf{w} \leftarrow (\mathbf{w}, b)$, so $f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x})$.

Signed distance from $\mathbf{x}$ to the hyperplane $\mathbf{w}^\top \mathbf{x} + b = 0$

Decompose $\mathbf{x} = \mathbf{x}_\perp + d \frac{\mathbf{w}}{\|\mathbf{w}\|}$ with $\mathbf{x}_\perp$ on the hyperplane. Then $\mathbf{w}^\top \mathbf{x} + b =$

$$\underbrace{\mathbf{w}^\top \mathbf{x}_\perp + b}_0 + d \frac{\mathbf{w}^\top \mathbf{w}}{\|\mathbf{w}\|} = d \|\mathbf{w}\|, \text{ so}$$

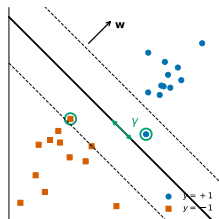
$$d = \frac{\mathbf{w}^\top \mathbf{x} + b}{\|\mathbf{w}\|}.$$

Margin

Geometric margin of an example, and of a dataset

$$\gamma_i = \frac{y_i(\mathbf{w}^\top \mathbf{x}_i + b)}{\|\mathbf{w}\|}, \quad \gamma = \min_i \gamma_i.$$

- $\gamma_i > 0$ iff $\mathbf{x}_i$ is on the correct side; its magnitude is the distance to the boundary.
- The data are **linearly separable** iff some $(\mathbf{w}, b)$ achieves $\gamma > 0$.
- γ is invariant to rescaling $(\mathbf{w}, b) \rightarrow (c\mathbf{w}, cb)$: it is a property of the hyperplane, not of its parametrization.



The Perceptron Algorithm (Rosenblatt, 1958)

Input: $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with $y_i \in \{-1, +1\}$; $\mathbf{x}_i$ augmented with a constant 1

```
w  $\leftarrow$  0;  
repeat  
  for  $i = 1, \dots, N$  do  
    if  $y_i \mathbf{w}^\top \mathbf{x}_i \leq 0$  ; // mistake  
      then  
         $\mathbf{w} \leftarrow \mathbf{w} + y_i \mathbf{x}_i$ ;  
      end  
    end  
until no mistakes in a full pass;  
Output: w
```

A mistake on $(\mathbf{x}_i, y_i)$ moves $\mathbf{w}$ toward $y_i \mathbf{x}_i$. After the update,

$$y_i (\mathbf{w} + y_i \mathbf{x}_i)^\top \mathbf{x}_i = y_i \mathbf{w}^\top \mathbf{x}_i + \|\mathbf{x}_i\|^2,$$

so the score on that example increases by $\|\mathbf{x}_i\|^2$. It need not become correct in one step.

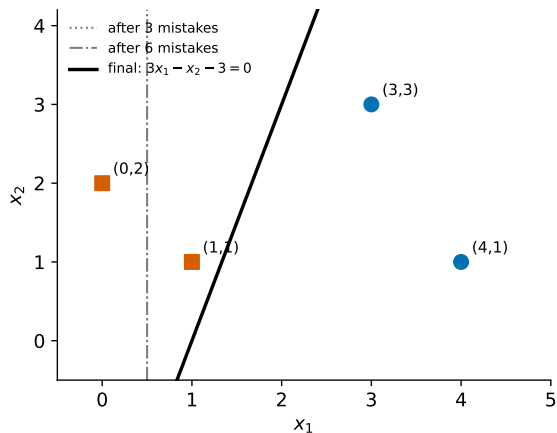
Numerical Example: Four Points, Seven Mistakes

Augmented inputs $\tilde{\mathbf{x}} = (x_1, x_2, 1)$, visited in the order shown, $\mathbf{w}^{(0)} = \mathbf{0}$. A score of exactly 0 counts as a mistake.

pass	$\tilde{\mathbf{x}}$	y	$\mathbf{w}^\top \tilde{\mathbf{x}}$	mistake?	new $\mathbf{w}$
1	(1, 1, 1)	-1	0	yes	(-1, -1, -1)
1	(3, 3, 1)	+1	-7	yes	(2, 2, 0)
1	(0, 2, 1)	-1	4	yes	(2, 0, -1)
1	(4, 1, 1)	+1	7	no	
2	(1, 1, 1)	-1	1	yes	(1, -1, -2)
2	(3, 3, 1)	+1	-2	yes	(4, 2, -1)
2	(0, 2, 1)	-1	3	yes	(4, 0, -2)
2	(4, 1, 1)	+1	14	no	
3	(1, 1, 1)	-1	2	yes	(3, -1, -3)
3	(3, 3, 1), (0, 2, 1), (4, 1, 1)		3, -5, 8	no	
4	all four		-1, 3, -5, 8	no	converged

Final separator: $3x_1 - x_2 - 3 = 0$. Seven updates. $R^2 = \max_i \|\tilde{\mathbf{x}}_i\|^2 = 19$.

Numerical Example: The Picture



The Perceptron Mistake Bound (Novikoff, 1962)

Theorem

Suppose $\|\mathbf{x}_i\| \leq R$ for all i , and there exists $\mathbf{w}^*$ with $\|\mathbf{w}^*\| = 1$ and $y_i \mathbf{w}^{*\top} \mathbf{x}_i \geq \gamma > 0$ for all i (the data are separable with margin γ). Then the perceptron makes at most

$$\frac{R^2}{\gamma^2}$$

mistakes, regardless of the order in which examples are presented, and regardless of N .

The proof tracks two quantities across mistakes: how much $\mathbf{w}$ aligns with $\mathbf{w}^*$ (grows at least linearly), and how long $\mathbf{w}$ is (grows at most like a square root). They cannot both hold forever.

Mistake Bound: Proof

Let $\mathbf{w}_k$ be the weight vector after the k -th mistake, $\mathbf{w}_0 = \mathbf{0}$, and let the k -th mistake be on $(\mathbf{x}, y)$.

Lower bound: alignment with $\mathbf{w}^*$ grows by at least γ per mistake

$$\mathbf{w}^{*\top} \mathbf{w}_k = \mathbf{w}^{*\top} (\mathbf{w}_{k-1} + y\mathbf{x}) = \mathbf{w}^{*\top} \mathbf{w}_{k-1} + \underbrace{y\mathbf{w}^{*\top} \mathbf{x}}_{\geq \gamma} \geq \mathbf{w}^{*\top} \mathbf{w}_{k-1} + \gamma \implies \mathbf{w}^{*\top} \mathbf{w}_k \geq k\gamma.$$

Upper bound: the norm grows by at most R^2 per mistake

$$\|\mathbf{w}_k\|^2 = \|\mathbf{w}_{k-1} + y\mathbf{x}\|^2 = \|\mathbf{w}_{k-1}\|^2 + 2 \underbrace{y\mathbf{w}_{k-1}^\top \mathbf{x}}_{\leq 0 \text{ (a mistake)}} + \underbrace{y^2 \|\mathbf{x}\|^2}_{\leq R^2} \leq \|\mathbf{w}_{k-1}\|^2 + R^2 \implies \|\mathbf{w}_k\|^2 \leq kR^2.$$

Combine with Cauchy-Schwarz

$$k\gamma \leq \mathbf{w}^{*\top} \mathbf{w}_k \leq \|\mathbf{w}^*\| \|\mathbf{w}_k\| = \|\mathbf{w}_k\| \leq \sqrt{k} R \implies k \leq \frac{R^2}{\gamma^2}. \quad \square$$

Reading the Bound

- The bound does not depend on N or on the dimension p . It depends on the geometry only: how spread out the data are (R) relative to how separable they are (γ).
- It is a bound on *mistakes*, not on passes. Correct examples cost nothing.
- It says nothing about which separator you end up with — only that you end up with one. Different orders give different $\mathbf{w}$. The SVM (next section) picks a canonical one.

If the data are not separable, the perceptron never terminates: it cycles forever. In practice one runs a fixed number of epochs and keeps the $\mathbf{w}$ with the best validation error — an early-stopping rule.

The perceptron update is SGD with $\eta = 1$ on the *perceptron loss* $\max(0, -y\mathbf{w}^\top \mathbf{x})$, whose subgradient at a mistake is $-\mathbf{y}\mathbf{x}$. This loss is convex but has a trivial minimizer ($\mathbf{w} = \mathbf{0}$), which is why the algorithm is defined by its updates rather than by its objective.

Support Vector Machines

Which Separator?

When data are separable, infinitely many hyperplanes achieve zero training error.

Maximum-margin principle

Choose the hyperplane whose geometric margin $\gamma = \min_i \gamma_i$ is largest.

Two reasons:

- Robustness: the classifier's predictions on the training set survive the largest perturbation of the inputs.
- Generalization: bounds on test error for linear classifiers scale with R^2/γ^2 — the same quantity as the perceptron's mistake bound. A large margin is evidence of a simple problem.

The maximizing hyperplane is unique, and it is determined by the few points that attain the minimum margin — the **support vectors**.

From Margin to a Quadratic Program

Fix the scale, then the objective becomes $\|\mathbf{w}\|$

Maximize γ subject to every point having margin at least γ :

$$\max_{\mathbf{w}, b, \gamma} \gamma \quad \text{s.t.} \quad \frac{y_i(\mathbf{w}^\top \mathbf{x}_i + b)}{\|\mathbf{w}\|} \geq \gamma \quad \forall i.$$

The constraints are invariant to rescaling $(\mathbf{w}, b)$. Use that freedom to impose $\min_i y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$. Then $\gamma = 1/\|\mathbf{w}\|$, and maximizing $1/\|\mathbf{w}\|$ is minimizing $\|\mathbf{w}\|$, or equivalently $\frac{1}{2}\|\mathbf{w}\|^2$:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 \quad \forall i$$

- A convex quadratic objective with linear inequality constraints: a **quadratic program**. Unique solution, solvable by off-the-shelf methods.
- The constraint $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$ is the *functional* margin; the geometric margin is $1/\|\mathbf{w}\|$.

The Lagrangian and the Support Vectors

Introduce multipliers $\alpha_i \geq 0$ for the constraints:

$$\mathcal{L}(\mathbf{w}, b, \alpha) = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^N \alpha_i [y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1].$$

Stationarity in $\mathbf{w}$ and b

$$\nabla_{\mathbf{w}} \mathcal{L} = \mathbf{w} - \sum_i \alpha_i y_i \mathbf{x}_i = \mathbf{0} \Rightarrow \boxed{\mathbf{w} = \sum_{i=1}^N \alpha_i y_i \mathbf{x}_i}, \quad \frac{\partial \mathcal{L}}{\partial b} = -\sum_i \alpha_i y_i = 0 \Rightarrow \sum_i \alpha_i y_i = 0.$$

Complementary slackness

At the optimum, $\alpha_i [y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1] = 0$ for every i . So either $\alpha_i = 0$, or the constraint is tight: $y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$, i.e. $\mathbf{x}_i$ sits exactly on the margin.

$\mathbf{w}$ is a linear combination of the training points, with nonzero weight only on those on the margin. Delete any other point and the solution does not change.

Soft Margin

Real data are not separable. Allow each constraint to be violated by a slack $\xi_i \geq 0$, and pay for it:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

Eliminate the slacks: the constraint problem is an unconstrained loss

For fixed $(\mathbf{w}, b)$, the optimal ξ_i is the smallest feasible one: $\xi_i = \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))$. Substituting,

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b))$$

This is empirical risk minimization with the **hinge loss** $\ell_{\text{hinge}}(y, f) = \max(0, 1 - yf)$ and an ℓ_2 penalty on $\mathbf{w}$ — the recipe of Lecture 1 again, plus the regularizer of Lecture 3. C trades margin width against training violations.

The Hinge Loss

Hinge is a convex upper bound on 0/1

For $y \in \{-1, +1\}$ and score f , let $m = yf$ (the functional margin).

- $\ell_{0/1} = \mathbf{1}\{m \leq 0\}$ and $\ell_{\text{hinge}} = \max(0, 1 - m)$.
- If $m \leq 0$: $1 - m \geq 1 = \ell_{0/1}$. If $m > 0$: $\ell_{\text{hinge}} \geq 0 = \ell_{0/1}$. So $\ell_{\text{hinge}} \geq \ell_{0/1}$ always.
- $\max(0, 1 - m)$ is the maximum of two affine functions of m , hence convex in m , hence convex in $\mathbf{w}$.

The Hinge Loss: Subgradient

Subgradient

ℓ_{hinge} is not differentiable at $m = 1$. A subgradient with respect to $\mathbf{w}$ is

$$\partial_{\mathbf{w}} \ell_{\text{hinge}} = \begin{cases} -y\mathbf{x} & \text{if } y(\mathbf{w}^{\top} \mathbf{x} + b) < 1, \\ \mathbf{0} & \text{if } y(\mathbf{w}^{\top} \mathbf{x} + b) > 1, \end{cases}$$

and anything in between at equality. SGD with this subgradient is the standard large-scale SVM solver.

Minimizing an upper bound on the 0/1 risk controls the 0/1 risk. That is the whole logic of surrogate losses.

Logistic Regression

Modelling a Probability

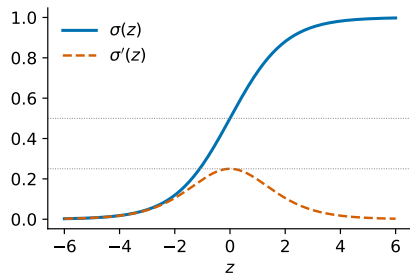
Instead of a hard label, model $p(\mathbf{x}) = \mathbb{P}(Y = 1 \mid \mathbf{x})$ for $y \in \{0, 1\}$. A linear function is the wrong object: $\mathbf{x}^\top \boldsymbol{\beta}$ ranges over $\mathbb{R}$. Pass it through the **logistic (sigmoid) function**:

$$\sigma(z) = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}, \quad p(\mathbf{x}) = \sigma(\mathbf{x}^\top \boldsymbol{\beta}).$$

Log-odds are linear

$$\frac{p}{1-p} = \frac{e^z / (1 + e^z)}{1 / (1 + e^z)} = e^z \Rightarrow \log \frac{p(\mathbf{x})}{1 - p(\mathbf{x})} = \mathbf{x}^\top \boldsymbol{\beta}.$$

A unit increase in x_j multiplies the odds by e^{β_j} .



The Derivative of the Sigmoid

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

$$\sigma'(z) = \frac{d}{dz}(1 + e^{-z})^{-1} = -(1 + e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2} = \underbrace{\frac{1}{1 + e^{-z}}}_{\sigma(z)} \cdot \underbrace{\frac{e^{-z}}{1 + e^{-z}}}_{1 - \sigma(z)}.$$

Also useful: $1 - \sigma(z) = \sigma(-z)$, and $\sigma'(z) \leq \frac{1}{4}$ with equality at $z = 0$.

The identity $\sigma' = \sigma(1 - \sigma)$ is the reason every gradient in this section is short. It reappears in ML II as the source of the vanishing-gradient problem.

Likelihood

The model says $Y_i \mid \mathbf{x}_i \sim \text{Bernoulli}(p_i)$ with $p_i = \sigma(\mathbf{x}_i^\top \boldsymbol{\beta})$, independently. The probability of the observed labels is

$$L(\boldsymbol{\beta}) = \prod_{i=1}^N p_i^{y_i} (1 - p_i)^{1-y_i}.$$

Log-likelihood

$$\begin{aligned} \ell(\boldsymbol{\beta}) &= \log L(\boldsymbol{\beta}) = \sum_{i=1}^N \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right] \\ &= \sum_{i=1}^N \left[y_i \log \frac{p_i}{1 - p_i} + \log(1 - p_i) \right] = \sum_{i=1}^N \left[y_i \mathbf{x}_i^\top \boldsymbol{\beta} - \log(1 + e^{\mathbf{x}_i^\top \boldsymbol{\beta}}) \right], \end{aligned}$$

using $\log \frac{p}{1-p} = \mathbf{x}^\top \boldsymbol{\beta}$ and $1 - p_i = \frac{1}{1 + e^{\mathbf{x}_i^\top \boldsymbol{\beta}}}$.

The MLE is $\hat{\boldsymbol{\beta}} = \arg \max_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta})$. Equivalently we minimize $-\ell$, the **log loss**.

The Gradient

$$\nabla_{\beta} \ell(\beta) = \sum_{i=1}^N (y_i - p_i) \mathbf{x}_i$$

Differentiate term by term, with $z_i = \mathbf{x}_i^{\top} \beta$ and $\nabla_{\beta} z_i = \mathbf{x}_i$:

$$\nabla_{\beta} \left[y_i z_i - \log(1 + e^{z_i}) \right] = y_i \mathbf{x}_i - \frac{e^{z_i}}{1 + e^{z_i}} \mathbf{x}_i = y_i \mathbf{x}_i - \sigma(z_i) \mathbf{x}_i = (y_i - p_i) \mathbf{x}_i.$$

- Same shape as least squares: residual $(y_i - p_i)$ times input. The residual is now a difference of a label and a probability.
- Setting the gradient to zero gives the **score equations** $\sum_i (y_i - p_i) \mathbf{x}_i = \mathbf{0}$: nonlinear in β through p_i , hence no closed form.
- With an intercept, the first score equation reads $\sum_i y_i = \sum_i p_i$: the fitted probabilities sum to the observed count. (Compare: least-squares residuals sum to zero.)

The Hessian, and Concavity

$$\nabla_{\beta}^2 \ell(\beta) = - \sum_{i=1}^N p_i(1 - p_i) \mathbf{x}_i \mathbf{x}_i^{\top}$$

Differentiate the gradient once more. Only $p_i = \sigma(\mathbf{x}_i^{\top} \beta)$ depends on β :

$$\nabla_{\beta} [(y_i - p_i) \mathbf{x}_i] = -\mathbf{x}_i (\nabla_{\beta} p_i)^{\top} = -\mathbf{x}_i (\sigma'(z_i) \mathbf{x}_i)^{\top} = -p_i(1 - p_i) \mathbf{x}_i \mathbf{x}_i^{\top}.$$

The Hessian is negative semidefinite, so ℓ is concave

For any $\mathbf{v}$,

$$\mathbf{v}^{\top} \nabla^2 \ell \mathbf{v} = - \sum_{i=1}^N p_i(1 - p_i) \underbrace{(\mathbf{v}^{\top} \mathbf{x}_i)^2}_{\geq 0} \leq 0,$$

since $p_i(1 - p_i) \in (0, \frac{1}{4}]$. Strictly negative when $\mathbf{X}$ has full column rank.

Concave log-likelihood $\Rightarrow$ convex log loss $\Rightarrow$ every stationary point is the global MLE, and gradient descent (with $L = \frac{1}{4} \lambda_{\max}(\mathbf{X}^{\top} \mathbf{X})$) converges at the $O(1/t)$ rate of §1.

Newton's Method

Gradient descent uses the first-order Taylor expansion. Newton's method uses the second:

Maximize the quadratic model

$$\ell(\boldsymbol{\beta} + \boldsymbol{\delta}) \approx \ell(\boldsymbol{\beta}) + \nabla \ell(\boldsymbol{\beta})^\top \boldsymbol{\delta} + \frac{1}{2} \boldsymbol{\delta}^\top \nabla^2 \ell(\boldsymbol{\beta}) \boldsymbol{\delta}.$$

Setting the $\boldsymbol{\delta}$ -gradient to zero: $\nabla \ell + \nabla^2 \ell \boldsymbol{\delta} = \mathbf{0}$, so

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - [\nabla^2 \ell(\boldsymbol{\beta}^{(t)})]^{-1} \nabla \ell(\boldsymbol{\beta}^{(t)}).$$

- The Hessian is invertible (negative definite) whenever $\mathbf{X}$ has full column rank, so the step is well defined.
- Near the optimum, Newton converges *quadratically*: the number of correct digits doubles per iteration. Five to ten iterations typically suffice.
- Cost per iteration $O(Np^2 + p^3)$ — the same as one least-squares solve. For p in the thousands, use gradient methods instead.

Newton = Iteratively Reweighted Least Squares

Write $\mathbf{p} = (p_1, \dots, p_N)^\top$ and $\mathbf{W} = \text{diag}(p_i(1 - p_i))$. In matrix form,

$$\nabla \ell = \mathbf{X}^\top (\mathbf{y} - \mathbf{p}), \quad \nabla^2 \ell = -\mathbf{X}^\top \mathbf{W} \mathbf{X}.$$

Rewrite the Newton step as a weighted least-squares solve

$$\begin{aligned} \beta^{(t+1)} &= \beta^{(t)} + (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{y} - \mathbf{p}) \\ &= (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \left[\mathbf{X}^\top \mathbf{W} \mathbf{X} \beta^{(t)} + \mathbf{X}^\top (\mathbf{y} - \mathbf{p}) \right] = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \underbrace{\left[\mathbf{X} \beta^{(t)} + \mathbf{W}^{-1} (\mathbf{y} - \mathbf{p}) \right]}_{\mathbf{z}} \\ &= \boxed{(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{z}} \end{aligned}$$

Each Newton step is the solution of $\min_{\beta} \sum_i W_{ii} (z_i - \mathbf{x}_i^\top \beta)^2$: a least-squares problem on the *working response* $\mathbf{z}$, with weights that are largest for uncertain points ($p_i \approx \frac{1}{2}$). Weights and working response are recomputed each iteration — hence **IRLS**.

Numerical Example: One IRLS Step

Three points with intercept: $\mathbf{x}_i = (1, x_i)$, $x = (-1, 0, 1)$, $y = (0, 0, 1)$. Start at $\boldsymbol{\beta}^{(0)} = \mathbf{0}$.

Step

At $\boldsymbol{\beta} = \mathbf{0}$ every $p_i = \sigma(0) = \frac{1}{2}$, so $\mathbf{W} = \frac{1}{4}\mathbf{I}$ and $\mathbf{z} = \mathbf{X}\mathbf{0} + \mathbf{W}^{-1}(\mathbf{y} - \mathbf{p}) = 4(\mathbf{y} - \frac{1}{2}\mathbf{1}) = (-2, -2, 2)^\top$.

$$\mathbf{X}^\top \mathbf{W} \mathbf{X} = \frac{1}{4} \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{W} \mathbf{z} = \frac{1}{4} \begin{pmatrix} -2 - 2 + 2 \\ 2 + 0 + 2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix}, \quad \boldsymbol{\beta}^{(1)} = \begin{pmatrix} 4/3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} \\ 2 \end{pmatrix}.$$

Check against the raw Newton form: $\nabla \ell(\mathbf{0}) = \sum_i (y_i - \frac{1}{2}) \mathbf{x}_i = (-\frac{1}{2}, 1)^\top$ and $-\nabla^2 \ell(\mathbf{0}) = \frac{1}{4} \mathbf{X}^\top \mathbf{X}$, so $\boldsymbol{\beta}^{(1)} = \mathbf{0} + (\frac{1}{4} \mathbf{X}^\top \mathbf{X})^{-1} (-\frac{1}{2}, 1)^\top = (-\frac{2}{3}, 2)^\top$. Same answer.

After the step, $p = \sigma(-\frac{2}{3} - 2, -\frac{2}{3}, -\frac{2}{3} + 2) \approx (0.065, 0.339, 0.791)$; the score equations are not yet zero, and three more steps bring $\boldsymbol{\beta}$ to convergence.

When the MLE Does Not Exist

If the training data are linearly separable, logistic regression has **no** maximum-likelihood estimate.

Why

Let β_0 separate the data: $y_i = 1 \Rightarrow \mathbf{x}_i^\top \beta_0 > 0$ and $y_i = 0 \Rightarrow \mathbf{x}_i^\top \beta_0 < 0$. Consider $\beta = c\beta_0$ with $c \rightarrow \infty$. Then $p_i \rightarrow 1$ for every $y_i = 1$ and $p_i \rightarrow 0$ for every $y_i = 0$, so each term of ℓ increases toward 0 and $\ell(c\beta_0) \uparrow 0 = \sup \ell$. The supremum is approached only as $\|\beta\| \rightarrow \infty$; it is never attained.

- Numerically: coefficients blow up, fitted probabilities saturate at 0 and 1, standard errors are meaningless.
- The fix is to penalize $\|\beta\|$ — Lecture 3. Regularized logistic regression always has a unique solution.
- Contrast with the perceptron, for which separable data are the *good* case.

Asymptotics

Under standard regularity conditions the MLE is consistent and asymptotically normal:

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathcal{I}(\boldsymbol{\beta}_0)^{-1}), \quad \mathcal{I}(\boldsymbol{\beta}_0) = \mathbb{E}[p(\mathbf{x})(1 - p(\mathbf{x})) \mathbf{x}\mathbf{x}^\top].$$

- $\mathcal{I}$ is the Fisher information — the negative expected Hessian per observation. Its sample version is $\frac{1}{N}\mathbf{X}^\top \mathbf{W}\mathbf{X}$, the matrix IRLS inverts at convergence.
- Standard errors are the square roots of the diagonal of $(\mathbf{X}^\top \hat{\mathbf{W}}\mathbf{X})^{-1}$; Wald, score and likelihood-ratio tests follow.
- The parallel with least squares is exact: there, $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$; here, $\text{Var}(\hat{\boldsymbol{\beta}}) \approx (\mathbf{X}^\top \mathbf{W}\mathbf{X})^{-1}$, with the variance of Y_i absorbed into $W_{ii} = p_i(1 - p_i)$.

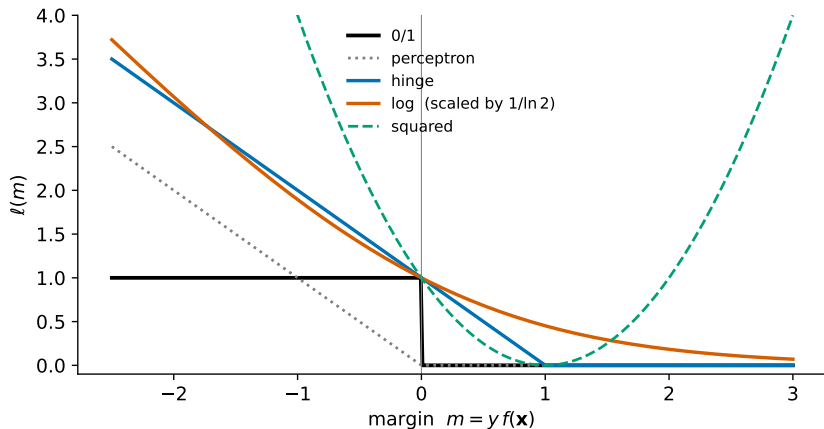
The Loss Functions, Side by Side

One Axis: The Margin

Convert logistic regression to $y \in \{-1, +1\}$: with $\tilde{y} = 2y - 1$, one checks $-\log p_i^{y_i}(1 - p_i)^{1-y_i} = \log(1 + e^{-\tilde{y}_i \mathbf{x}_i^\top \beta})$. Every loss today is a function of the margin $m = yf(\mathbf{x})$:

Loss	$\ell(m)$	Minimizer, and what it estimates
0/1	$\mathbf{1}\{m \leq 0\}$	the Bayes classifier; intractable
Perceptron	$\max(0, -m)$	any separator; degenerate on non-separable data
Hinge	$\max(0, 1 - m)$	max-margin separator; a <i>classifier</i> , not a probability
Log	$\log(1 + e^{-m})$	$\log \frac{P(y=1 \mathbf{x})}{P(y=-1 \mathbf{x})}$; a calibrated probability
Squared	$(1 - m)^2$	penalizes $m > 1$ too; wrong shape for classification

The Picture



Hinge and log both upper-bound 0/1, are convex, and vanish (or nearly) for confident correct predictions. Squared loss rises again on the right: it punishes being *too* right.

What We Proved Today

1. Descent lemma for L -smooth f ; with $\eta \leq 1/L$, $f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \frac{\eta}{2} \|\nabla f\|^2$.
2. Convex + smooth $\Rightarrow f(\mathbf{x}_t) - f^* \leq L \|\mathbf{x}_0 - \mathbf{x}^*\|^2 / (2t)$.
3. Perceptron: at most R^2/γ^2 mistakes on data separable with margin γ (Novikoff).
4. Max margin $\Leftrightarrow \min \frac{1}{2} \|\mathbf{w}\|^2$ s.t. $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$; $\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$ with $\alpha_i \neq 0$ only on support vectors; soft margin $\Leftrightarrow$ hinge loss + ℓ_2 penalty.
5. Logistic regression: $\nabla \ell = \mathbf{X}^\top (\mathbf{y} - \mathbf{p})$, $\nabla^2 \ell = -\mathbf{X}^\top \mathbf{W} \mathbf{X} \preceq 0$; Newton = IRLS; no MLE on separable data.

Next

Lecture 3 (Monday) — ridge and lasso: closed forms, shrinkage in the SVD basis, why ℓ_1 gives exact zeros; then evaluation: precision, recall, ROC, and cost-sensitive thresholds.

Lab 2 (Wednesday, Rodrigo) — ridge vs. lasso paths, cross-validated λ , and the separable-data failure of unregularized logistic regression, seen live.

Reading — *ESL* §4.4 (logistic regression), §4.5 (separating hyperplanes), §12.1–12.2 (SVM). Novikoff (1962) is two pages and worth reading in the original.